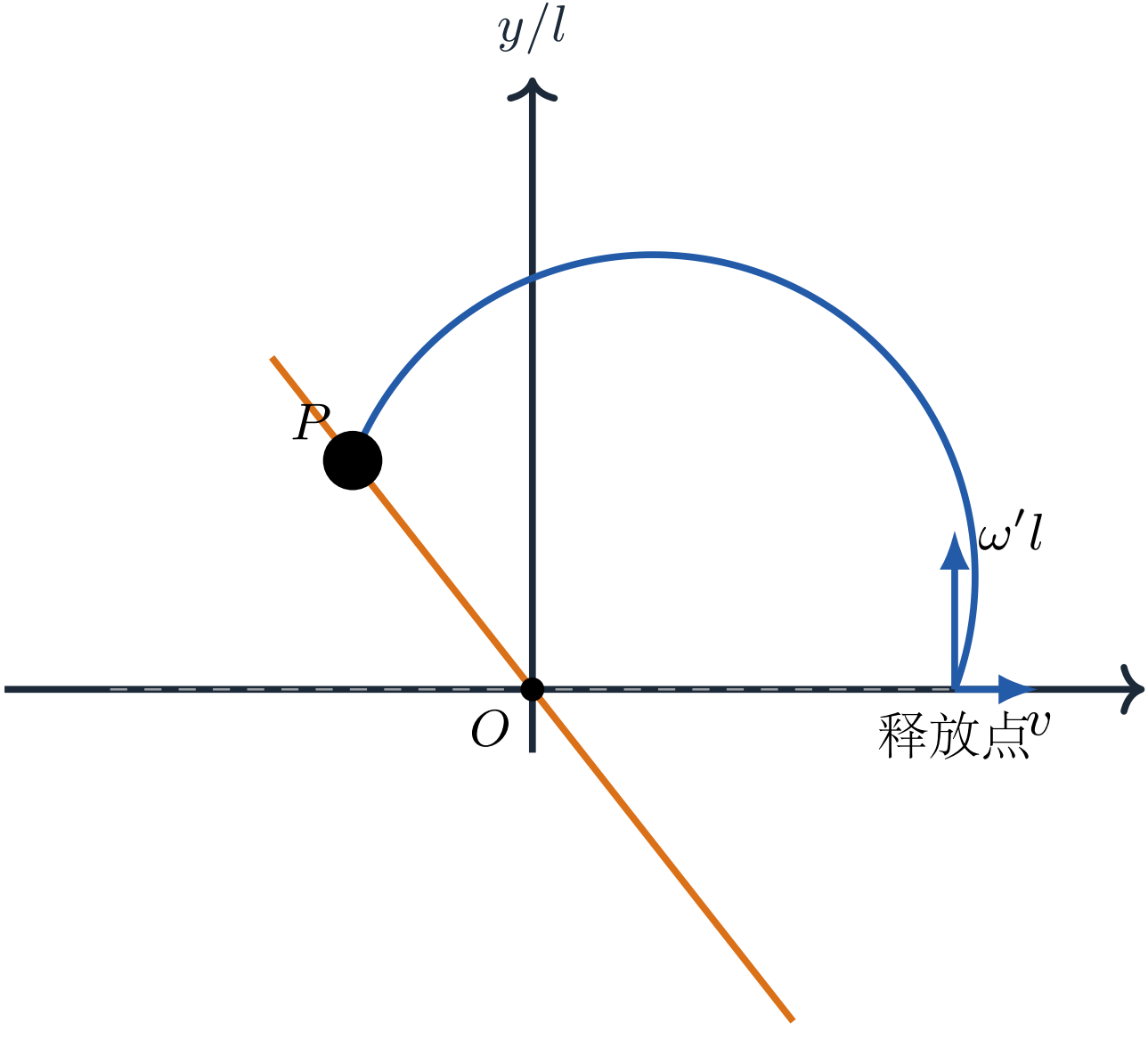

$\alpha = 1$

$\varphi_P = 2.236$ rad

$s_P/l = 0.6894$

$F'(\tau_P) = 0.1631 \neq 0$

蓝色：自由小环；橙色：相交时的有限杆段